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Summary

Computer Science graduate with experience across machine learning, computer vision, and full-stack Python development. Built and shipped three real internship projects: a PyTorch leaf disease detection system with a custom attention model, a Django web app with four live API integrations, and a face recognition attendance system at an IOC refinery. Comfortable working across deep learning pipelines, relational databases, and REST APIs, depending on what the problem needs.

Experience

Intern - ATEES Global

Nov 2025 – Jun 2026

Data Science Intern

Thrissur, Kerala

- Designed AttCM-AlexNet from scratch in PyTorch, a custom CNN with channel, spatial, and self-attention, to classify plant leaf diseases from a single image.
- Deployed the full inference pipeline on Streamlit: upload → 224x224 preprocessing → prediction with confidence score, end-to-end in one flow.
- Pulled real-time weather data via REST API to generate disease-specific remedies and advisories tailored to current field conditions.

Intern - Grapesgenix Technical Solutions Pvt Ltd.

Jun 2025 – July 2025

Full-Stack Web Developer Intern

Thrissur, Kerala

- Built MyMovieList, a Django web app where users track, rate, and review movies, anime, series, and comics, i.e., all on one platform.
- Wired in four external APIs (TMDB, Jikan, Kitsu, Comic Vine) simultaneously to fetch live posters, ratings, and metadata per content type.
- Handled user auth end-to-end: registration & login, session management, password hashing, and a custom profile dashboard.
- Modelled the SQLite database with Django ORM using content inheritance to cleanly support multiple media categories without redundant tables.

Intern - Indian Oil Corporation Limited

Dec 2024 – Jan 2025

Python Developer Intern

Vadodara, Gujarat

- Built a face recognition attendance system using OpenCV's LBPH algorithm to eliminate manual tracking and proxy entries during any live Toolbox Talk sessions at IOC Gujarat Refinery.
- Automated employee verification end-to-end, replacing manual roll calls and making proxy attendance practically impossible.
- Logged every attendance entry with timestamps into MySQL in real time; added a ReportLab export so HR could pull structured PDF/CSV reports directly.
- Designed the Tkinter desktop UI simple enough that floor workers could register, check logs, and navigate without any training.

Technical Skills

Languages: Python, C, C++, SQL

Frameworks & Web: Django, Streamlit, Tkinter, HTML5, CSS3, JavaScript, Bootstrap 5, Tailwind CSS

AI / ML / CV: PyTorch, TensorFlow, OpenCV, MediaPipe, Scikit-Learn, NumPy, Matplotlib

Libraries & Tools: Pillow, ReportLab, Torchvision, Pandas, REST APIs, Git, GitHub, VS Code, Google Colab

Databases: MySQL, SQLite3, Django ORM

Platforms: Windows, WSL2/Linux (Ubuntu)

Education

GSFC University, Vadodara, Gujarat, India

Oct 2022 - June 2026

Bachelor of Technology in Computer Science Engineering — CGPA: 7.16 / 10

Vadodara, Gujarat

Kendriya Vidyalaya No 1 AFS Jamnagar, Gujarat, India

July 2022

12th CBSE — Percentage: 77.60 / 100

Jamnagar, Gujarat

Projects

AgroScan: Plant Leaf Disease Detection & Advisory System | *PyTorch, Streamlit, OpenCV, REST API*

- Built AttCM-AlexNet from scratch in PyTorch, a modified AlexNet with Channel, Spatial, and Self-Attention modules plus residual connections. Hence, the model focuses on diseased leaf regions rather than background noise.
- Trained the model on a multi-class plant disease dataset with 224x224 image preprocessing, batch normalisation, dropout regularisation, and a softmax classification head.
- Wired a complete inference pipeline in Streamlit: image upload, tensor preprocessing, disease prediction, and confidence score output running in one continuous flow.
- Fetched live weather data through a REST API and used temperature and humidity readings to generate disease-specific remedies and field advisories tied to actual environmental conditions.

Real-Time Dynamic Sign Language Detector (In Development) | *Python, WSL2, VS Code*

- Extracting 126 spatial hand landmarks per frame using MediaPipe and computing velocity vectors and relative positional offsets to make predictions consistent regardless of hand size or camera distance.
- Designed a sequence model with a Bidirectional LSTM (128 units) paired with a Temporal Attention layer to classify gestures across a rolling 30-frame window rather than isolated hand positions.
- Collecting 30+ samples per sign (hello, thank you, help, yes, no) with a custom data recording script that saves raw landmark sequences as CSVs for preprocessing and training.
- Running GPU-accelerated training on an RTX 4050 via WSL2 with CUDA, targeting continuous real-time detection without button triggers between signs.

MyMovieList: Multi-Platform Entertainment Tracker | *Python, Django, SQLite3, JavaScript, REST APIs*

- Built the full platform using Django's MVT pattern with user registration, login, session management, password hashing, and a custom profile dashboard - all from scratch.
- Connected four external APIs (TMDB, Jikan, Kitsu, Comic Vine) in parallel to pull live posters, ratings, and metadata across movies, series, anime, and comics into a single interface.
- Designed a normalised SQLite schema with Django ORM using a content inheritance model so movies, series, anime, and comics share a clean base structure without duplicating tables.
- Built watchlist management with status tracking (Planning, Watching, Completed, On Hold, Dropped), user ratings, reviews, and a helpful votes system.
- Kept the frontend responsive across mobile and desktop using Bootstrap 5 and JavaScript with Django templates handling dynamic content rendering.

Attendance Management System using Face Recognition | *Python, OpenCV, MySQL, Tkinter, Pillow*

- Built a desktop attendance tool using OpenCV's LBPH algorithm for real-time face recognition, to eliminate manual tracking and proxy entries during any live Toolbox Talk sessions at IOC Gujarat Refinery.
- Piped recognition output directly into MySQL, logging each verified entry with a timestamp and replacing what was previously a fully manual roll call process.
- Added a ReportLab export module so HR administrators could extract the structured PDF and CSV attendance reports without touching the database directly.
- Designed the Tkinter interface simple enough for floor workers to register faces, check logs, and navigate the system without any prior training.